MODULE 17 / LESSON 01

AI Agents and LangGraph Workflows

Coordinate stateful steps and bounded tool use with explicit graph control.

DSACADEMY

WHY THIS MATTERS

Start from what you already know

LLM applications, structured output, tools, retries, state, and evaluation

01 Explain workflows and agents in plain language.

02 Use state nodes and edges in a small worked example.

03 Complete the guided lab and verify checkpoints and stopping conditions.

CORE CONCEPT

The vocabulary of this topic

Coordinate stateful steps and bounded tool use with explicit graph control.

workflows and agents: A workflow follows declared control flow; an agent chooses actions dynamically from observations and goals.

state nodes and edges: State stores the run record, nodes transform it, and edges define allowed transitions.

tools and observations: A tool performs a bounded operation; its returned observation becomes evidence for the next decision.

checkpoints and stopping conditions: Checkpoints persist resumable state; stopping conditions bound attempts, steps, time, and cost.

FOUR CONNECTED IDEAS

How the concept works

01

workflows and agents →

02

state nodes and edges →

03

tools and observations →

04

checkpoints and stopping conditions

State enters a node, the node returns an update, edges choose the next node, and explicit limits bound execution.

CORE CONCEPT

Read every part of the mechanism

State enters a node, the node returns an update, edges choose the next node, and explicit limits bound execution.

state: The typed data carried through a run.

node: One transformation of state.

edge: A permitted transition.

checkpoint: Saved resumable state.

WORKED EXAMPLE

Worked example by hand

Route a ticket: billing goes to lookup, technical goes to diagnostics, and unknown goes to human review.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
def route(state):
    return {"billing": "lookup", "technical": "diagnose"}.get(
        state["category"], "human_review"
    )
```

OUTPUT

```
billing -> lookup
```

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model

- [x] Use deterministic workflows when steps are known; grant an agent choice only where it adds value.
- [x] Typed state makes every node input, output, and routing decision inspectable.
- [x] Tools need schemas, authorization, timeouts, resource limits, and observable results.
- [x] Bound total steps and attempts, checkpoint recoverable state, and route uncertainty to a human.

FOUR CONNECTED IDEAS

Notebook readiness map

01

workflows and agents

→

02

state nodes and edges

→

03

tools and observations

→

04

checkpoints and stopping conditions

Use the concepts from this lesson to read and complete https://github.com/curiousily/AI-Bootcamp/blob/master/29.langgraph-quickstart.ipynb.

RISK REVIEW

Common beginner mistakes

01 Using an open-ended agent when deterministic routing is sufficient.

02 Allowing a graph to revisit a node without a step or attempt limit.

GUIDED LAB

Build a typed LangGraph workflow with conditional routing, one tool, a checkpoint, and a step limit.

01

Predict

02

Run

03

Inspect

04

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

01 **Explain workflows and agents in plain language.**

02 **Use state nodes and edges in a small worked example.**

03 **Complete the guided lab and verify checkpoints and stopping conditions.**

NEXT EVIDENCE

A tested graph with state schema, rendered routes, tool permissions, execution traces, and failure recovery.